

The COMS Randomized Trial of Iodine 125 Brachytherapy for Choroidal Melanoma

IV. Local Treatment Failure and Enucleation in the First 5 Years after Brachytherapy. COMS Report No. 19

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Objective: To describe the frequency and predictors of local treatment failure and enucleation after iodine 125 (¹²⁵) brachytherapy in patients with choroidal melanoma treated and followed up in a large randomized clinical trial.

Design: Prospective, noncomparative, interventional case series within a randomized, multicenter clinical trial.

Participants: Patients enrolled in the Collaborative Ocular Melanoma Study (COMS) trial of enucleation versus brachytherapy between February 1987 and July 1998; tumors measured 2.5 to 10.0 mm in apical height and no more than 16.0 mm in longest basal dimension.

Methods: ¹²⁵ brachytherapy was administered via episcleral plaque according to a standard protocol. Follow-up ophthalmic evaluations, including ophthalmic ultrasound and fundus photography, were performed according to a standard protocol at baseline, every 6 months thereafter for 5 years, and subsequently at annual intervals. Survival analysis methods were used to estimate the cumulative risk of postirradiation treatment failure and enucleation. Factors associated with treatment failure and enucleation of plaquet eyes were evaluated using Cox proportional hazards analysis.

Main Outcome Measures: Reports of enucleation and of local treatment failure, defined as tumor growth, recurrence, or extrascleral extension, derived from clinical reports based on echographic and photographic documentation.

Results: As of September 30, 2000, 638 of the 650 patients randomized to brachytherapy and so treated had been followed up for 1 year or longer, and 411 had been followed up for at least 5 years. Sixty-nine eyes were enucleated during the first 5 years after brachytherapy, and treatment failure was reported for 57 eyes. The Kaplan-Meier estimate of proportion of patients undergoing enucleation by 5 years was 12.5% (95% confidence interval [CI], 10.0%–15.6%); the risk of treatment failure was 10.3% (95% CI, 8.0%–13.2%). Treatment failure was the most common reason for enucleation within 3 years of treatment; beyond 3 years, ocular pain was most common. Risk factors for enucleation were greater tumor thickness, closer proximity of the posterior tumor border to the foveal avascular zone, and poorer baseline visual acuity in the affected eye. Risk factors for treatment failure were older age, greater tumor thickness, and proximity of the tumor to the foveal avascular zone. Local treatment failure was associated weakly with reduced survival after controlling for baseline tumor and personal characteristics (adjusted risk ratio, 1.5; *P* = 0.08).

Conclusions: Local treatment failure and enucleation were relatively infrequent events after ¹²⁵ brachytherapy within the COMS. Treatment failure typically occurred early and was associated weakly with poorer survival. The COMS randomized trial documented the absence of a clinically or statistically significant difference in survival for patients randomly assigned to enucleation versus brachytherapy. This analysis documents the efficacy of brachytherapy to achieve sustained local tumor control and to conserve the globe. *Ophthalmology* 2002;109:2197–2206 © 2002 by the American Academy of Ophthalmology.

The goal of treatment for choroidal melanoma is to preserve the patient's life through removal or destruction of the tumor, traditionally achieved by enucleation of the involved

eye or by radiation therapy.^{1,2} The Collaborative Ocular Melanoma Study (COMS) has conducted a randomized trial to evaluate the effectiveness of globe-conserving radiother-

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apy (iodine 125 [I¹²⁵] brachytherapy) with respect to patient survival.³ Ocular brachytherapy is targeted to control local tumor (i.e., eliminate metastatic potential) while preserving the globe and potentially maintaining visual function (visual field, visual acuity, or both). The COMS Group recently reported survival outcomes for patients undergoing either enucleation or brachytherapy⁴; similar survival rates were found in the treatment arms through 12 years of follow-up. Those results strengthen the ongoing evolution of one of the ocular oncologist's treatment approaches for selected tumors: to achieve local tumor control through the application of globe-conserving therapy. Episcleral plaque radiotherapy (brachytherapy) is now the most commonly applied treatment in the management of medium-sized choroidal melanoma.^{1,2} Previously, there have been no reports of brachytherapy treatment failure based on a large, prospective, multicenter, clinical case series. The COMS patient cohort provides a unique opportunity to evaluate treatment efficacy and morbidity in a large, well-characterized group of patients for whom treatment was assigned randomly and brachytherapy was administered in a standardized manner.

For the purpose of this report, local treatment failure (hereafter referred to as treatment failure) was defined as local intraocular tumor growth, recurrence, or extraocular extension. Treatment failure presumes active tumor growth and the potential for development of metastatic disease leading to death. Enucleation for reasons other than treatment failure (typically severe ocular pain or uncontrollable neovascular glaucoma) also was evaluated in the context of

Table 1. Scheduled Follow-up Examinations Expected and Completed during 5 Years after Enrollment for Patients Assigned to and Receiving Iodine 125 Brachytherapy in the Collaborative Ocular Melanoma Study

Time since Enrollment (mos)	Patients Receiving Plaque (n = 650)	
	Expected* (No.)	Completed (No. (%))
6	647	638 (99)
12	638	617 (97)
24	623	570 (91)
36	558	507 (91)
48	484	419 (87)
60	411	346 (84)

* Number of patients assigned to and administered iodine 125 brachytherapy who had not been reported as deceased.

brachytherapy failure. Although enucleation for such indications would not be considered a predisposition to metastatic disease in the absence of viable tumor, it clearly represents a failure to achieve the secondary goal of globe conservation. Previous evaluations of I¹²⁵ brachytherapy in the treatment of choroidal melanoma typically have included patients treated at a single institution with potential limitations in terms of patient characteristics, definition of outcomes, and data collection and analysis methods.⁵⁻¹⁶ This report documents the incidence of treatment failure and secondary enucleation occurring within the COMS randomized trial among patients treated at 43 centers within North America and establishes a framework to discuss these outcomes with patients who have comparable tumor characteristics and may be considering globe-conserving I¹²⁵ brachytherapy.

Materials and Methods

Patients and Data Collection

The COMS study design has been described in a previous report.³ Briefly, the COMS Group conducted two randomized, controlled clinical trials designed to evaluate the role of radiotherapy in the treatment of choroidal melanoma. The trial of interest for this analysis compared I¹²⁵ brachytherapy with enucleation for tumors measuring at least 2.5 mm but no more than 10.0 mm in apical height and no more than 16.0 mm in longest basal dimension. Eligible patients had a primary choroidal melanoma present in only one eye, were at least 21 years of age, were judged to be suitable candidates for either brachytherapy or enucleation, had no other primary or metastatic malignancy other than carcinoma in situ of the uterine cervix or noninvasive nonmelanotic skin cancers, had no coexisting disease judged likely to compromise 5-year survival, were not receiving immunosuppressive therapy, and had no social, mental, or physical condition that would preclude their return for treatment and follow-up. Patients were ineligible if the tumor could not be delineated clearly by echograms, could not be observed adequately because of insufficient clarity of the ocular media, was diffuse or had more than one focus, was contiguous with the optic disc, was within 2 mm of the optic disc and was judged by the ophthalmologist to be unable to be plaqued satisfactorily, was predominantly located in the ciliary body, involved

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A complete listing of the COMS Group as of September 30, 2000, may be found in *Arch Ophthalmol* 2001;119:969-82.

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Table 2. Baseline Characteristics of Patients Treated by Brachytherapy in the Collaborative Ocular Melanoma Study

Characteristic	No.	%
Demographic characteristics		
Gender		
Male	325	50.0
Female	325	50.0
Age (yrs)		
<50	163	25.1
50–69	327	50.3
≥70	160	24.6
Race		
White	638	98.2
Other	12	1.8
Ophthalmic characteristics		
Visual acuity in eye with tumor		
≥20/20	211	32.5
20/25–20/40	244	37.5
20/50–20/160	126	19.4
≤20/200	63	9.7
NA	6	0.9
Visual acuity in fellow eye		
≥20/20	466	71.7
20/25–20/40	162	24.9
20/50–20/160	19	2.9
NA	3	0.5
Lens status in eye with tumor		
Phakic	609	93.7
Aphakic or pseudophakic	41	6.3
Tumor characteristics		
Tumor apical height (mm)		
2.5–5.0	419	64.5
5.1–7.5	167	25.7
7.6–10.0	64	9.8
Longest basal dimension (mm)		
4.5–8.0	87	13.4
8.1–11.0	222	34.2
11.1–14.0	237	36.5
14.1–16.0	104	16.0
Location of anterior tumor border		
Ciliary body	72	11.1
Equator to ora serrata	224	34.5
Posterior to equator	354	54.5
Location of posterior border		
Equator to ora serrata	17	2.6
Posterior to equator	633	97.4
Retina detached over tumor		
Yes	364	56.0
No	286	44.0
Distance to proximal edge of optic disc (mm)		
≤2.0	96	14.8
2.1–4.0	201	30.9
4.1–6.0	158	24.3
6.1–8.0	96	14.8
>8.0	97	14.9
NA	2	0.3
Distance to center of foveal avascular zone (mm)		
0.0	87	13.4
0.1–2.0	163	25.1
2.1–5.0	189	29.1
5.1–8.0	107	16.5
>8.0	98	15.1
NA	6	0.9
Tumor shape		
Dome	502	77.2
Collar button	104	16.0
Other	44	6.8

(continued)

Table 2. (continued)

Characteristic	No.	%
Treatment characteristics		
Dose to tumor apex (Gy)		
≤85.0	159	24.5
85.1–98.0	165	25.4
98.1–120.0	159	24.5
≥120.0	159	24.5
NA	8	1.2
Dose to sclera (Gy)		
<293.0	316	48.6
293.0–342.9	148	22.8
343.0–409.9	119	18.3
≥410	67	10.3
NA = data not available.		

the angle or iris, had extrascleral extension of more than 2 mm, or had been treated or biopsied previously. Fellow eye visual acuity of 20/200 or worse, diagnosis of neovascular or secondary glaucoma in the eye with the tumor, or prior treatment for a condition secondary to the melanoma also rendered a patient ineligible.

Patients gave signed informed consent before enrollment and randomization to either enucleation or brachytherapy. The institutional review boards of each of the participating institutions approved the COMS protocol and consent forms used at the respective clinical centers before the initiation of enrollment at the individual centers.

All patients with medium tumors were evaluated according to a standard protocol,¹⁷ with all information reported on standard data collection forms.¹⁸ An extensive medical history and ophthalmic examination were performed at baseline, every 6 months thereafter for 5 years, and once each year thereafter. The ophthalmic examination included measurement of best-corrected visual acuity, ophthalmoscopy, examination of the tumor-containing eye and fellow eye, and gonioscopy when indicated. Standardized A-scan and contact B-scan echography were performed by a COMS-certified echographer at baseline and at each follow-up examination.^{19,20} Color fundus photography was performed at the baseline evaluation and at each follow-up visit; fluorescein angiography was performed at the baseline, 2-year, 5-year, and 8-year examinations. Echograms were evaluated centrally at an echography center (Mars Hill, NC), and photographs and angiograms were evaluated at a photograph reading center (Iowa City, IA) according to standard grading protocols. A cancer-oriented physical examination, chest radiograph, and liver function tests were performed at baseline and at annual follow-up examinations, with abnormal findings assessed by diagnostic imaging or biopsy. Telephone interviews were conducted when a patient missed a scheduled clinic examination and, after the fifth annual examination, once each year between annual examinations. The telephone interview included assessment of current health, vision status in both eyes, and information on medical care received from ophthalmologists or other physicians since the previous examination or telephone interview. All study data were sent to the COMS Coordinating Center (Baltimore, MD) for entry into a central database.

If treatment failure was identified during follow-up examination by the following ophthalmologist, a separate form documenting details of the diagnosis was completed and submitted to the Coordinating Center. In the event that a plaqued eye was enucleated, an additional report was submitted. Whenever the initial brachytherapy was deemed to have failed or a plaqued eye was scheduled for enucleation, echograms and photographs were submitted to the respective reading centers. Eyes enucleated during

Table 3. Primary Reason for Enucleation of Plaque Eye within 5 Years of Collaborative Ocular Melanoma Study Enrollment

Reason for Enucleation*	Time to Enucleation						Total	
	<6 mos		6 mos–3 yrs		>3 yrs			
	n	%	n	%	n	%	n	%
Local treatment failure	5	83	25	61	9	41	39	57
Pain	0	—	9	22	10	45	19	28
Pain and low VA	0	—	1	2	1	5	2	3
Low VA	1	17	3	7	2	9	6	9
Other								
Retinal detachment	0	—	1	2	0	—	1	1
Suspected tumor growth	0	—	2	5	0	—	2	3
Total	6	100	41	100	22	100	69	100

VA = visual acuity.

* Reason reported at time of enucleation surgery; three patients with history of treatment failure had the eye enucleated for other reasons.

follow-up were processed according to the COMS pathology protocol^{17,21} by COMS-certified pathologists at the clinical centers or the central pathology center (Madison, WI). Stained sections of the tumor were reviewed independently by each member of the Pathology Review Committee. Cell type, coded using the modification of the Callendar classification developed at the Armed Forces Institute of Pathology,²² was determined. After adjudicated review, detailed histopathologic findings were reported to the clinical center and to the Coordinating Center.

Between the initiation of enrollment in February 1987 and the end of accrual in July 1998, 1317 patients were enrolled. Of these, 657 were randomized to I¹²⁵ brachytherapy. Seven patients (1.1%) who did not receive the protocol brachytherapy as assigned were excluded from this analysis. The remaining 650 patients are the focus of this report.

Iodine 125 Brachytherapy Protocol

Plaque size was chosen so that the tumor base plus a tumor-free margin of 2 to 3 mm was covered entirely by the plaque. The

tumor base was defined to include all pigmented areas. Notched or trimmed plaques were acceptable for tumors located within 2 mm of the optic nerve, with the edge of the plaque placed between the margin of the tumor and the optic nerve. The prescription point was defined as the tumor apex for tumors measuring 5.0 mm or more in apical height; but for smaller tumors, the prescription point was 5 mm from the interior surface of the sclera. In 1996, the COMS adopted the recommendations of the American Association of Physics in Medicine Task Group (TG-43) for dosimetry of interstitial sources in anticipation of a revised calibration standard for I¹²⁵ seeds by the National Institute for Standards and Technology. Under the TG-43 formalism, the absorbed dose at the prescription point was 85 Gy delivered at the rate of at least 0.42 Gy/hour, but no more than 1.05 Gy/hour.^{23,24} Before adopting the TG-43 formalism, the prescribed total dose was calculated to be 100 Gy; however, the actual amount of radiation delivered was not affected by the change. All dose data included in this report were recalculated based on the TG-43 formalism. Actual radiation dose delivered to the prescription point, tumor apex, sclera at the center of the plaque, and critical structures within the eye was reported to

Table 4. Kaplan-Meier Estimates of Time to Local Treatment Failure and Time to Enucleation in Patients Treated with Iodine 125 Brachytherapy

Interval since enrollment (mos)	Number at Risk	Number of Events during Interval	Cumulative Percentage with Event	95% Confidence Interval
Time to local treatment failure				
0–6	650	14	2.2	1.3–3.6
7–12	613	4	2.8	1.8–4.4
13–18	591	9	4.3	3.0–6.2
19–24	561	4	5.0	3.5–7.0
25–30	535	9	6.6	4.9–8.8
31–36	483	7	7.9	6.0–10.4
37–42	432	2	8.3	6.4–10.9
43–48	408	3	9.0	6.9–11.7
49–54	368	4	10.0	7.8–12.9
55–60	331	1	10.3	8.0–13.2
Time to enucleation				
0–6	650	7	1.1	0.5–2.3
7–12	631	7	2.2	1.3–3.7
13–18	614	12	4.1	2.8–6.0
19–24	594	7	5.2	3.7–7.3
25–30	563	6	6.2	4.6–8.4
31–36	518	8	7.7	5.8–10.1
37–42	472	5	8.7	6.7–11.2
43–48	436	4	9.5	7.4–12.2
49–54	400	9	11.5	9.1–14.5
55–60	361	4	12.5	10.0–15.6

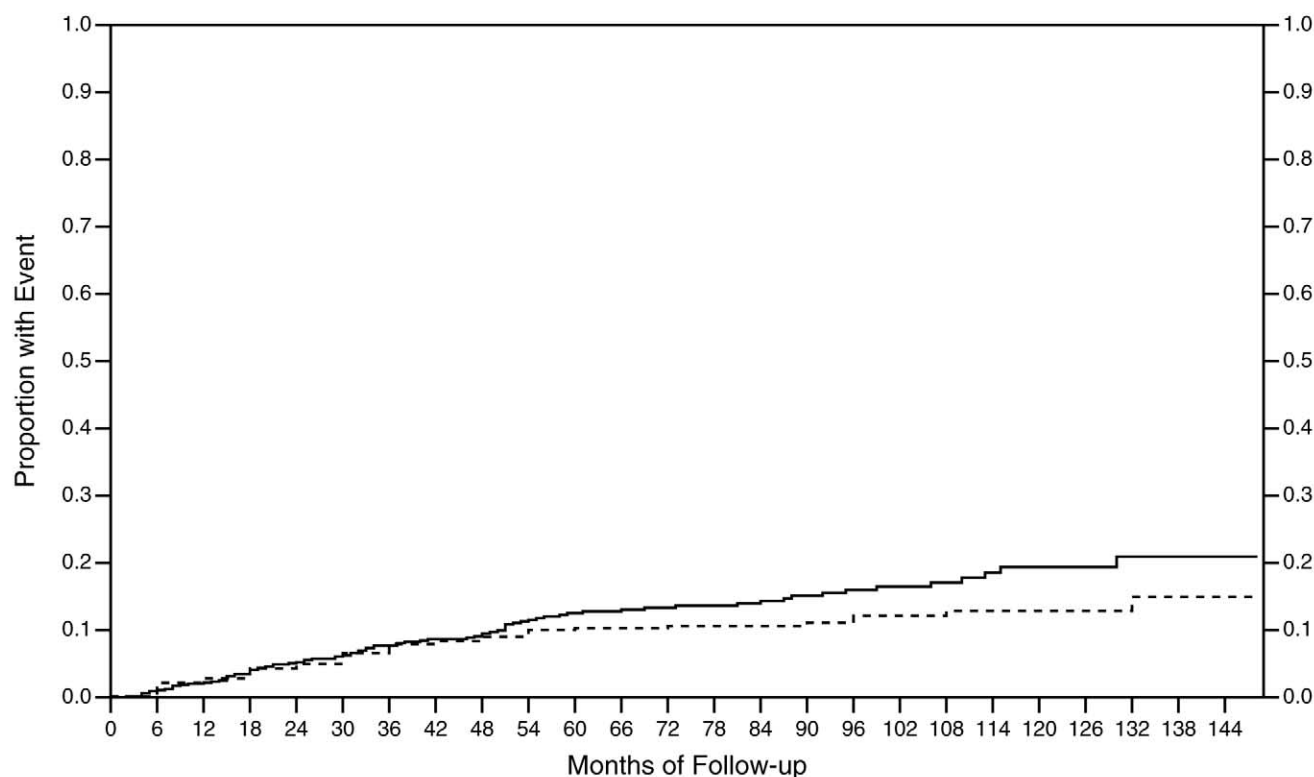


Figure 1. Cumulative proportion of patients undergoing secondary enucleation (solid line) or with reported local treatment failure (dashed line) after iodine 125 brachytherapy. See Table 4 for corresponding numbers of events and persons at risk for each interval.

and confirmed by the COMS Radiologic Physics Center (Houston, TX). Doses were calculated based on presumed plaque location, I^{125} seed activity and location in the plaque, and times of plaque insertion and removal.

Outcome Measures

For this report, measures of local treatment failure were based on the clinical determination of tumor expansion or extrascleral extension greater than 2 mm after brachytherapy. Protocol criteria for tumor growth were based on a two-stage observation of estimated tumor dimensions: (1) an initial increase in height of at least 15% as measured by echography or a 250- μ m expansion of any tumor boundary as judged from photographs or clinical examination; (2) an additional 15% increase in elevation or a further 250- μ m expansion of any tumor boundary observed and confirmed on subsequent examination. When growth exceeding the specified minimum was observed, the ophthalmologist did not always wait to confirm this at a second visit before initiating further treatment (usually enucleation). Extrascleral extension was based on clinical or echographic findings. In addition, local recurrence (orbital mass, ciliary body mass, retinal invasion) in the absence of documented tumor expansion or extrascleral extension was included as treatment failure for this analysis.

Statistical Methods

All analyses were based on data received at the Coordinating Center by September 30, 2000. As of that date, all patients had been enrolled in the COMS for at least 2 years and most had been followed up for at least 5 years. Frequency distributions of the

baseline demographic, ophthalmic, and tumor characteristics and selected treatment and follow-up characteristics were constructed to describe the patients. For the time to event analyses, outcome was defined as the date of the first report of documented treatment failure or the date of enucleation. Life table analysis with the log-rank test²⁵ was used to identify characteristics that were associated with the outcomes. Intervals for the analysis correspond to the COMS follow-up visit schedule, which required examinations every 6 months after enrollment, with an allowable 6-month window centered on a target date based on the anniversary of the date of enrollment. For the Kaplan-Meier analysis of time to enucleation,²⁶ patients were censored as of their last clinical examination or telephone interview. For time to treatment failure, patients were censored as of the date of enucleation or, for those retaining their eye, date of last clinical examination.

Characteristics associated with time to treatment failure or enucleation with statistical significance of $P < 0.20$ by the log-rank test were included in a multivariate analysis using the Cox proportional hazards model,²⁷ which tests each characteristic for association with time to outcome while adjusting for other characteristics included in the model. A backward stepwise procedure that included variables that were significantly associated with the outcome at a statistical significance level of $P \leq 0.10$ was used to select the best model.

To investigate whether treatment failure or secondary enucleation of an eye treated initially with brachytherapy was associated with poorer survival outcome, separate Cox proportional hazards models were evaluated for each outcome. The models incorporated initially only time to enucleation or time to reported treatment failure as a time-dependent covariate; baseline variables found to be associated with time to death in the entire cohort of COMS

Table 5. Cumulative Percentages of Patients with Local Treatment Failure within 5 Years, Stratified by Baseline and Follow-up Characteristics

Characteristic	No. with Event	% with Event*
Demographic characteristics		
Gender		
Male	27	10.6
Female	30	10.7
Age (yrs) [‡]		
<50	8	5.7
50–69	34	12.6
≥70	15	11.8
Ophthalmic characteristics		
Visual acuity in eye with tumor [‡]		
≥20/20	12	6.8
20/25–20/40	23	11.0
20/50–20/160	17	18.5
≤20/200	5	8.5
Tumor and treatment characteristics		
Apical height (mm) [†]		
2.5–5.0	29	7.9
5.1–7.5	20	15.2
7.6–10.0	8	20.2
Largest basal dimension (mm) [†]		
4.5–8.0	4	4.8
8.1–11.0	20	10.3
11.1–14.0	18	10.0
14.1–16.0	15	19.2
Distance to proximal edge of optic disc (mm)		
≤2	9	10.8
2.1–4.0	22	12.8
4.1–6.0	11	8.6
6.1–8.0	10	12.4
>8.0	4	5.7
Distance to foveal avascular zone (mm) [‡]		
0.0	13	17.4
0.1–2.0	15	10.3
2.1–5.0	17	10.7
5.1–8.0	3	3.9
>8.0	8	10.2
Location of anterior border		
Ciliary body	6	12.0
Equator to ora serrata	16	8.2
Posterior to equator	35	11.8
Location of posterior border		
Equator to ora serrata	0	0
Posterior to equator	57	10.8
Retina detached over tumor [‡]		
No	18	7.6
Yes	39	13.0
Tumor shape		
Dome	11	10.0
Collar button	43	15.4
Other	3	7.5
Dose to tumor apex (Gy) [†]		
≤85	19	17.0
86–98	19	13.4
99–120	11	8.0
>120	8	5.8
Dose to sclera (Gy)		
<293.0	27	10.4
293.0–342.9	11	8.3
343.0–409.9	13	12.8
≥410	6	13.7

* Estimate of cumulative percentage with clinical report of local treatment failure at 5 years.

[†] $P < 0.05$, log-rank test.

[‡] $P < 0.10$, log-rank test.

patients in the trial of enucleation versus brachytherapy⁴ (age, apical height of the tumor, longest basal diameter, distance of the tumor to the optic disc, and tumor shape) were then added as covariates.

Results

Description of Patient Population and Treatment

As of September 30, 2000, the 650 patients assigned to and treated by I¹²⁵ brachytherapy had been followed up clinically for a median of 67 months (range, 0–153 months). No clinical follow-up was available for patients who either died before the first scheduled follow-up examination (three patients) or did not return for any examinations (seven patients). Six hundred thirty-eight patients (98%) were followed up clinically for at least 1 year, 623 (96%) for at least 2 years, and 411 (63%) for 5 years or more. The proportion of patients eligible for follow-up who completed the ophthalmic evaluation in each annual interval is displayed in Table 1. The 5-year clinical examination was completed by 346 of the 411 patients (84%) followed up for 5 or more years.

Patient demographic, tumor, and treatment characteristics are described in Table 2. Fifty percent of the population was male; patient ages at the time of COMS enrollment ranged from 24 to 88 years, with a median of 61 years. More than 98% of the patients were non-Hispanic white persons. Tumor height at baseline ranged from 2.5 to 10.0 mm, with a median of 4.2 mm. Largest basal dimension ranged from 4.5 to 16.0 mm, with a median of 11.5 mm. Fifty-five percent of the tumors were located entirely posterior to the equator; only 3% had a posterior tumor border located anterior to the equator. Fifteen percent of tumors were within 2.0 mm of the optic disc, with a median distance from the posterior tumor border to the optic disc of 4.5 mm. Three fourths of the tumors were dome shaped, 16% had a collar-button configuration, and a small number had other shapes, including peaked, lobulated, irregular, and flat. More than half of the patients (56%) had a retinal detachment over the tumor at baseline. Seventy percent had visual acuity in the tumor-containing eye of 20/40 or better; only 10% had 20/200 or worse visual acuity. In the fellow eye, 97% had visual acuity of 20/40 or better; patients with visual acuity of 20/200 or worse in the fellow eye were not eligible for the trial. Median visual acuities in the tumor-containing eye and fellow eye were 20/32 and 20/20, respectively.

Events during Follow-up

During the first 5 years after enrollment, 69 plaqued eyes were enucleated. Treatment failure was reported for 57 eyes. Most eyes with tumors exhibiting treatment failure were enucleated, but 15 were not; in three of these cases, the patient was diagnosed with metastasis before or soon after the report of tumor growth, or died within a few months of the noted tumor expansion, and one case did go on to enucleation after the 5-year cutoff for this report. The remaining patients continued to be followed up, with additional local treatment administered at the discretion of the ophthalmologist. Reasons for enucleation are presented in Table 3. Most enucleated eyes contained growing tumors; a small proportion were enucleated because of pain, poor visual acuity, or other reasons. Two eyes were enucleated because of extrascleral extension; for one of these, tumor growth was also reported, and for the other, orbital recurrence was reported. Reasons for enucleation varied depending on the elapsed time since enrollment. More than 80% of eyes enucleated during the first 6 months were removed because of reported treatment failure, although almost 60% of eyes enucleated after the first 3 years were removed for other reasons.

Table 6. Multivariate Cox Proportional Hazards Model for Association of Baseline Variables with Time to Local Treatment Failure

Variable	Risk Ratio	95% Confidence Interval	Chi-square	P Value
Age at baseline (yrs)				
<50 (ref.)	1.0		8.3	0.02
50–69	2.8	(1.3–6.5)		
≥69	2.9	(1.1–7.1)		
Apical height (mm)				
2.5–5.0 (ref.)	1.0		8.2	0.004
5.1–10.0	2.4	(1.3–4.2)		
Distance of tumor border to foveal avascular zone center (mm)				
0.0 (ref.)	1.0		18.3	0.0004
0.1–2.0	0.6	(0.3–1.3)		
2.1–8.0	0.4	(0.2–0.8)		
≥8.0	0.5	(0.2–1.2)		

For a small number of enucleated cases, tumor growth was suspected but could not be confirmed because of vitreous hemorrhage or some other barrier to accurate delineation of tumor margins; in most of these, pain or reduced visual acuity was also present.

Kaplan-Meier estimates of time to reported treatment failure and time to enucleation are shown in Table 4. By 5 years after enrollment, 10.3% of plaqued eyes (95% confidence interval [CI], 8.0–13.2%) exhibited treatment failure. The 5-year cumulative risk of enucleation in this cohort was slightly higher: 12.5% (95% CI, 10.0–15.6%). Estimates of time to treatment failure and to enucleation are shown in Figure 1.

Predictors of Local Treatment Failure

Factors associated with 5-year risk of treatment failure in the univariate analysis (Table 5) included greater apical height and basal dimension of the tumor, retinal detachment over the tumor, and lower radiation dose to the tumor apex ($P < 0.05$ for all variables). Older age at enrollment, poorer visual acuity in the study eye at baseline, and closer proximity of the posterior tumor border to the foveal avascular zone (FAZ) were marginally associated with treatment failure ($P < 0.10$). In multivariate analysis of baseline predictors of time to reported treatment failure (Table 6), only older age at time of enrollment, greater apical height, and proximity to the FAZ were associated independently with treatment failure. The risk in persons over age 50 was nearly three times as high as in those less than 50 years of age.

Predictors of Enucleation

Risk factors for enucleation were evaluated separately. Univariate predictors of enucleation (<0.05) within 5 years (Table 7) included male gender, greater apical height of the tumor, longer basal dimension, poorer visual acuity in the tumor-containing eye at baseline, collar-button tumor shape, presence of retinal detachment over the tumor, lower radiation dose to the tumor apex, and higher dose to the sclera. Distance to the FAZ was marginally associated with enucleation ($P = 0.15$). In multivariate analysis adjusting for univariate predictors of enucleation, only apical height of the tumor, poorer baseline visual acuity in the tumor-containing eye, and distance between the proximal tumor border and the FAZ were independent predictors of outcome, although gender bordered on statistical significance (Table 8).

In a separate analysis of enucleation according to the primary

reason, men had nearly three times the risk of enucleation for reasons other than treatment failure compared with women (risk ratio, 2.8; 95% CI, 1.2–6.8). Greater tumor height and worse baseline vision in the study eye also were associated with enucleation for reasons other than treatment failure in multivariate analysis (data not shown). Only closer proximity of the posterior tumor border to the FAZ was associated independently with enucleation because of treatment failure (data not shown).

Local Treatment Failure and Enucleation as Predictors of Time to Death

Enucleation of the plaqued eye during follow-up was not an independent predictor of time to death in either univariate or multivariate analysis. Treatment failure, which was strongly associated with reduced survival in the univariate model (risk ratio, 1.9; $P = 0.01$), was only weakly associated after adjusting for other baseline variables (adjusted risk ratio, 1.5; $P = 0.08$; 95% CI, 0.94–2.52).

Histopathologic diagnosis was reviewed for eyes enucleated after brachytherapy. To determine whether tumor cell type was associated with increased risk of treatment failure, histopathologic diagnosis was compared for eyes enucleated because of treatment failure versus other reasons. Tumors were classified as being primarily of epithelioid, spindle, or mixed cell type. Among eyes enucleated because of treatment failure, 14% were classified as having epithelioid cell tumors, whereas only 3% of eyes enucleated for other reasons were so classified; however, the difference was not statistically significant. Overall, plaqued eyes enucleated during follow-up for any reason were significantly more likely to contain epithelioid cell tumors than eyes randomly assigned to enucleation and so treated initially (9% vs. 3%, chi-square, 22.4, 3 degrees of freedom; $P = 0.001$).

Discussion

Our present understanding of cure for a patient with choroidal melanoma requires local control of the tumor before dissemination of viable tumor cells elsewhere in the body. In the past, there has been much debate over whether dissemination of the tumor occurs before the institution of therapy, at the time of manipulation of the eye with institution of local therapy, or after definitive local therapy by enucleation, radiation, or other globe-sparing method.^{28,29} In the COMS, local control of the tumor with radioactive plaque was usually successful. We have defined failure of local plaque therapy to include cases where extrascleral extension of the tumor occurred after brachytherapy, continued growth of the tumor was documented, or the tumor initially responded to radiation and then recurred. These cases represent a failure of the radiation therapy to control the local tumor biologically. Enucleation of the eye also represents a failure of radiation therapy in that the aim of treatment is to retain the eye.

The COMS showed a very high rate of successful control of the tumor locally, with nearly 90% of patients exhibiting tumor control and 88% retaining the eye at 5 years. Treatment failure and enucleation as a result of treatment failure, pain, or other complications were relatively uncommon events in COMS patients treated with radioactive plaque. Comparison of our results with those of other investigations is difficult because of differences in study design, patient

Table 7. Cumulative Percentages of Patients with Enucleation of Plaque Eye within 5 Years, Stratified by Baseline and Follow-up Characteristics

Characteristic	No. with Event	% with Event*
Demographic characteristics		
Gender [§]		
Male	45	17.2
Female	24	8.7
Age (yrs)		
<50	15	10.5
50–69	39	14.1
≥70	15	12.4
Ophthalmic characteristics		
Visual acuity in eye with tumor [‡]		
≥20/20	11	6.6
20/25–20/40	32	15.7
20/50–20/160	16	15.9
≤20/200	9	16.9
Tumor characteristics		
Apical height (mm) [†]		
2.5–5.0	23	6.4
5.1–7.5	27	20.3
7.6–10.0	19	38.1
Largest basal dimension (mm) [*]		
4.5–8.0	6	7.3
8.1–11.0	20	11.0
11.1–14.0	26	13.7
14.1–16.0	17	20.8
Distance to proximal edge of optic disc (mm)		
≤2	13	15.9
2.1–4.0	24	14.8
4.1–6.0	14	11.1
6.1–8.0	10	12.0
>8.0	7	8.1
Distance to foveal avascular zone center (mm)		
0.0	14	18.8
0.1–2.0	21	15.8
2.1–5.0	16	9.7
5.1–8.0	8	10.3
>8.0	9	11.2
Location of anterior border		
Ciliary body	6	10.3
Equator to ora serrata	25	14.1
Posterior to equator	38	12.6
Location of posterior border		
Equator to ora serrata	1	6.7
Posterior to equator	68	12.9
Retina detached over tumor [§]		
No	18	7.6
Yes	51	17.1
Tumor shape [†]		
Dome	41	9.7
Collar button	23	30.2
Other	5	11.8
Treatment characteristics		
Dose to tumor apex (Gy) [†]		
≤85	27	21.6
86–98	27	19.7
99–120	7	5.1
>120	8	6.4
Dose to sclera (Gy) [*]		
<293.0	24	9.4
293.0–342.9	11	8.5
343.0–409.9	20	18.9
≥410	14	27.5

* Estimate of cumulative percentage enucleated at 5 years.

† $P < 0.001$, log-rank test.‡ $P < 0.01$, log-rank test.§ $P < 0.05$, log-rank test.|| $P < 0.20$, log-rank test.

selection, follow-up time, definition of growth or local failure, and methods of analysis. Most studies were retrospective clinical case series and most have presented the proportion of patients failing plaque therapy unadjusted for follow-up time; in addition, the criteria for plaque failure often were not defined clearly. Table 9 summarizes results from prior evaluations of I¹²⁵ brachytherapy for choroidal melanoma. The data cited in the table are based on the reported counts of events unadjusted for length of follow-up; nevertheless, wide variation in the rates of local plaque failure and enucleation is evident. The only published studies that presented life table estimates of plaque failure and secondary enucleation after I¹²⁵ brachytherapy showed higher failure rates than those seen in the COMS. Patients treated with I¹²⁵ brachytherapy in the report of Char et al¹¹ had a risk of enucleation of 8% at 2 years and 15% at 4 years. Quivey et al¹² reported an 18% 5-year risk of local treatment failure. Several other studies of brachytherapy for choroidal melanoma included patients treated with various isotopes, including I¹²⁵; however, it was not possible to distinguish the results for any particular radioisotope in the published reports.^{15,16} The effect of losses to follow-up in other studies is unknown. In the COMS, 5-year treatment failure and enucleation status was unknown for only 4% of the original cohort of 650 treated patients. Compared with the patients whose 5-year outcome status was known, the group with unknown status was older but otherwise similar with respect to baseline characteristics that were associated with treatment failure or enucleation. Thus, our results are unlikely to be biased as a result of differential response.

The 5-year overall mortality among patients assigned to radioactive plaque in the COMS was 18%.⁴ There was no obvious relationship between the performance of enucleation after brachytherapy and mortality in the first 5 years. However, multivariate analysis suggested that the presence of treatment failure was a weak risk factor for mortality in these patients, with a relative risk of mortality of 1.5 in patients with reported treatment failure compared with those for whom treatment failure was not reported ($P = 0.08$). These results are consistent with those of Egan et al³⁰ in patients followed up after treatment with proton beam radiotherapy. In that study, although enucleation for reasons other than tumor growth was not associated with reduced survival, there was a nearly fourfold increased risk of mortality for enucleation resulting from growth of the tumor.

The association between treatment failure and mortality may represent the effect of the continued presence of viable tumor cells in the eye, resulting either from location outside the radiation field or from resistance of the tumor to appropriate radiation treatment. The survival of viable tumor cells could foster dissemination and metastasis. Difficulty localizing or securing the plaque in some posteriorly located tumors adjacent to the optic nerve or FAZ could result in inadequate treatment to the tumor margin and consequent survival of viable tumor. To minimize the risk of inadequate radiation delivery to all tumor margins, care must be taken to localize and secure the plaque, especially in posterior tumors.

Another possibility, supported by our histopathologic findings, is that tumors showing recurrence after radiother-

Table 8. Multivariate Cox Proportional Hazards Model for Association of Baseline Variables with Enucleation of Plaque Eye within 5 Years

Variable	Risk Ratio	95% Confidence Interval	Chi-square	P Value
Gender				
Female (ref.)	1.0			
Male	1.6	(1.0–2.7)	3.5	0.06
Apical height (mm)				
2.5–5.0 (ref.)	1.0			
5.1–7.5	3.1	(1.7–5.6)	20.3	0.00004
7.6–10.0	5.0	(2.4–10.5)		
Visual acuity, tumor eye				
≥20/20 (ref.)	1.0			
20/25–20/40	2.0	(1.0–4.0)	16.3	0.001
20/50–20/160	1.6	(0.7–3.6)		
≤20/200	1.4	(0.5–3.7)		
Distance of tumor border to foveal avascular zone center (mm)				
0.0 (ref.)	1.0			
0.1–2.0	0.6	(0.3–1.3)	12.6	0.006
2.1–8.0	0.4	(0.2–0.9)		
≥8.0	0.5	(0.2–1.4)		

apy are more aggressive or dangerous tumors with a greater capacity to disseminate. In the COMS, because of randomization of treatment assignment, one would expect similar distributions of cell type among tumors in the brachytherapy arm and in the enucleation arm at baseline. However, tumors in eyes enucleated after brachytherapy were three times as likely to have been classified as epithelioid than the tumors of COMS patients randomly assigned to primary enucleation. These secondarily enucleated tumors (most with reported treatment failure) may have been more aggressive histologically to begin with, demonstrating a higher percentage of epithelioid cells than the average patient. Alternatively, treatment failure after brachytherapy may indicate the evolution of a more aggressive or dangerous cell type, that is, epithelioid. Finally, it should be considered that epithelioid cells may be more resistant to brachytherapy than spindle cells, thus being more likely to continue proliferating after therapy in all irradiated tumors.

Without an appropriate comparison group (i.e., plaque eyes not undergoing secondary enucleation), the significance of this observation cannot readily be determined.

Treatment failure was the primary reason for enucleation overall, but after 3 years, most eyes were enucleated because of pain and other factors. The risk factors for enucleation of the eye within the first 5 years included treatment failure, initial greater linear dimension, greater tumor thickness, and closer proximity of the posterior tumor border to the foveal avascular zone. Occurrence of treatment failure was associated similarly with greater tumor thickness, closer proximity of the posterior tumor border to the foveal avascular zone, poorer baseline visual acuity, and older age. Our results are similar to those reported by other investigators,^{12,13,15,16} although in their multivariate analysis, Quivey et al¹² found that smaller tumor height was a predictor for local tumor progression. Our observation of a weak association between enucleation risk and male gender has not to

Table 9. Tumor Control and Secondary Enucleation in Reports of Treatment of Choroidal Melanoma with Iodine 125 Brachytherapy

Report	No. of Patients	Mean Follow-up (mos)	Local Failure* (%)	Enucleation* (%)
Garretson et al ⁵	26	45	NA	8
Bosworth et al ⁶	58	49	14	14
Packer et al ⁷	64	65	8	17
Hill et al ⁸	21	73	29	24
Kreissig et al ⁹	19	60	11	0
Fontanesi et al ¹⁰	144	46	10	10
Char et al ^{11†}	98	41	13	17
Quivey et al ¹²	239	36	9	NA
Wilson and Hungerford ^{13‡}	190	47	4	6

NA = estimate not available in publication.

* Unadjusted percent of enrolled patients with local failure (varied definitions) or enucleation during follow-up (no. with events divided by total no. of patients).

† Total number studied, 184; results for the 98 patients treated with iodine 125 reported here.

‡ Total number studied for all isotopes, 597; results for the 190 patients treated with iodine 125 reported here.

our knowledge been reported previously. Because there was no association between gender and treatment failure, this finding, if real, may indicate higher risk in men of serious complications leading to enucleation, or it may reflect greater acceptance among men of enucleation after brachytherapy.

Our findings showing low risk of local treatment failure or secondary enucleation after definitive I¹²⁵ brachytherapy for choroidal melanoma, in combination with the lack of a survival difference between patients receiving primary enucleation and brachytherapy in the COMS,⁴ should lend further reassurance to ophthalmologists and patients considering I¹²⁵ brachytherapy for this disease. Although brachytherapy is a relatively safe procedure, the impact of radiation on visual acuity and vision-related functioning remain of concern. The COMS recently reported that nearly half of patients in this cohort experienced substantial loss of vision by 3 years after treatment (loss of six or more lines of visual acuity from the baseline level).³¹ The effect of treatment and vision loss on quality of life and visual functioning are being evaluated in a parallel study in this cohort.³²

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